

Core Java Interview Preparation Notes

1. Java Basics

- What is Java?
 - Features of Java
 - Platform Independent
 - Object-Oriented
 - Secure
 - Robust
 - Multithreaded
 - Portable
 - High Performance
 - JVM, JRE, JDK
 - Java Program Structure
 - Compilation and Execution Process
 - Bytecode
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2. Data Types and Variables

- Primitive Data Types
 - byte
 - short
 - int
 - long
 - float
 - double
 - char
 - boolean
 - Non-Primitive Data Types
 - Variable Types
 - Local Variable
 - Instance Variable
 - Static Variable
 - Type Casting
 - Implicit Casting
 - Explicit Casting
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3. Operators

- Arithmetic Operators
- Relational Operators
- Logical Operators
- Assignment Operators
- Unary Operators
- Bitwise Operators

- Ternary Operator
 - instanceof Operator
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4. Control Statements

Conditional Statements

- if
- if-else
- else-if ladder
- nested if
- switch

Looping Statements

- for loop
- while loop
- do-while loop
- enhanced for loop

Jump Statements

- break
 - continue
 - return
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5. Object-Oriented Programming (OOP)

Class and Object

- Class
- Object
- Object Creation

Constructors

- Default Constructor
- Parameterized Constructor
- Constructor Overloading

OOP Principles

Encapsulation

- Data Hiding
- Getters and Setters

Inheritance

- Single Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance

Polymorphism

- Method Overloading
- Method Overriding

Abstraction

- Abstract Class
- Abstract Methods

Interface

- Interface
 - Multiple Inheritance using Interface
 - Default Methods
 - Static Methods
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6. Keywords

- this
 - super
 - static
 - final
 - abstract
 - synchronized
 - volatile
 - transient
 - native
 - strictfp
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7. Arrays

- Single Dimensional Array
 - Multi-Dimensional Array
 - Array Initialization
 - Array Traversal
 - Array Class Methods
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8. String Handling

String Class

- Immutable Nature
- String Pool
- String Creation

String Methods

- length()
- charAt()
- substring()
- equals()
- equalsIgnoreCase()
- compareTo()
- replace()
- split()
- trim()
- contains()

StringBuilder

- Mutable
- append()
- insert()
- reverse()

StringBuffer

- Thread Safe
 - Difference between String, StringBuilder, StringBuffer
-

9. Wrapper Classes

- Integer
 - Double
 - Float
 - Character
 - Boolean
 - Autoboxing
 - Unboxing
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10. Exception Handling

Exception Hierarchy

- Throwable
- Error

- Exception

Types of Exceptions

- Checked Exception
- Unchecked Exception

Keywords

- try
- catch
- finally
- throw
- throws

Custom Exception

11. Collections Framework

List

- ArrayList
- LinkedList
- Vector
- Stack

Set

- HashSet
- LinkedHashSet
- TreeSet

Queue

- PriorityQueue
- Deque
- ArrayDeque

Map

- HashMap
- LinkedHashMap
- TreeMap
- Hashtable
- ConcurrentHashMap

Collection Concepts

- Comparable
- Comparator
- Iterator

- ListIterator
- Collections Utility Class

Interview Differences

- ArrayList vs LinkedList
 - HashMap vs Hashtable
 - HashMap vs ConcurrentHashMap
 - HashSet vs TreeSet
 - Comparable vs Comparator
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12. Java 8 Features

- Lambda Expressions
- Functional Interfaces
- Predicate
- Consumer
- Supplier
- Function
- Method References
- Stream API
- Optional Class
- Default Methods
- forEach()

Stream Operations

- filter()
 - map()
 - sorted()
 - distinct()
 - collect()
 - count()
 - reduce()
 - anyMatch()
 - allMatch()
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13. Multithreading

Thread Creation

- Extending Thread Class
- Implementing Runnable Interface

Thread Lifecycle

- New
- Runnable
- Running

- Waiting
- Blocked
- Terminated

Thread Methods

- start()
- run()
- sleep()
- join()
- yield()

Synchronization

- synchronized Method
- synchronized Block

Concepts

- Race Condition
- Deadlock
- Thread Safety

Executor Framework

- ExecutorService
- Callable
- Future

14. File Handling

File Class

- Create File
- Delete File
- Rename File

Streams

- FileInputStream
- FileOutputStream
- BufferedInputStream
- BufferedOutputStream

Readers/Writers

- FileReader
- FileWriter
- BufferedReader
- BufferedWriter

Serialization

- Serializable Interface
 - transient Keyword
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15. Java Memory Management

- Heap Memory
- Stack Memory
- Method Area
- Metaspace

Garbage Collection

- System.gc()
 - GC Types
 - finalize()
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16. Important Java Classes

- Object Class
 - Math Class
 - Arrays Class
 - Collections Class
 - Scanner Class
 - Date Class
 - LocalDate
 - LocalDateTime
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17. Access Modifiers

- private
 - default
 - protected
 - public
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18. Packages

- Built-in Packages
 - User-defined Packages
 - Import Statement
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19. Enum

- Enum Declaration
 - Enum Methods
 - Use Cases
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20. Important Interview Questions

JVM Related

- Difference between JVM, JRE, and JDK
- How Java achieves platform independence?
- What is ClassLoader?

OOP

- Explain all OOP principles.
- Difference between Abstract Class and Interface.
- Method Overloading vs Overriding.

String

- Why String is immutable?
- String Pool?
- String vs StringBuilder vs StringBuffer?

Collections

- Internal Working of HashMap.
- Why HashMap allows one null key?
- Difference between HashMap and Hashtable.

Exception

- Checked vs Unchecked Exception.
- throw vs throws.

Multithreading

- Runnable vs Thread.
- synchronized keyword.
- Deadlock.

Java 8

- Lambda Expression.
- Functional Interface.
- Stream API.
- Optional Class.

Memory

- Heap vs Stack.
 - Garbage Collection.
 - Memory Leak.
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Revision Checklist

- ✓ Java Basics
- ✓ OOP Concepts
- ✓ Strings
- ✓ Arrays
- ✓ Exception Handling
- ✓ Collections Framework
- ✓ Java 8 Features
- ✓ Multithreading
- ✓ File Handling
- ✓ Memory Management
- ✓ Access Modifiers
- ✓ Interview Questions